



LUMI AI Factory

Applications

Introduction to Quantum Computing 2026 – Essi Maaranen

lumi-ai-factory.eu

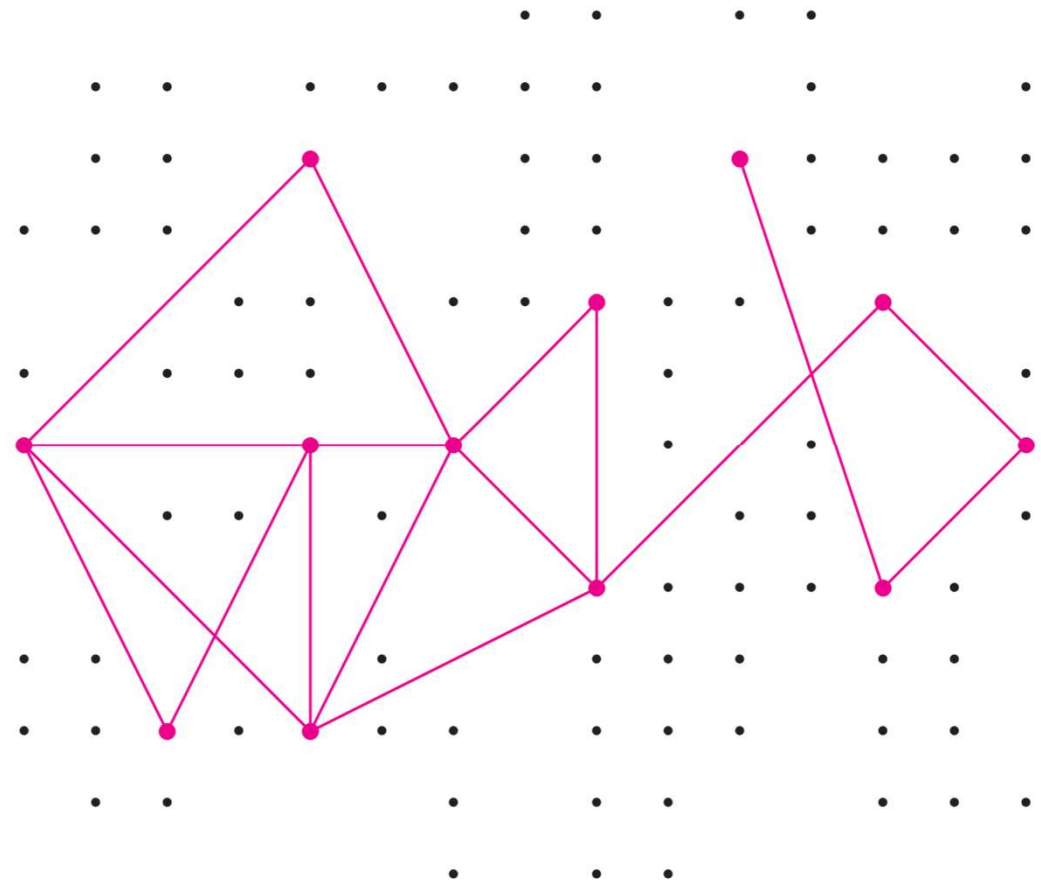
Overview

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3. Where Can Quantum Computers Offer Value
 1. Simulating Nature
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 3. Quantum Machine Learning
 4. Quantum Beyond Just Computing
4. Reality Check

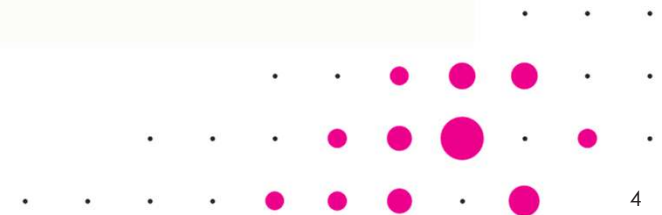
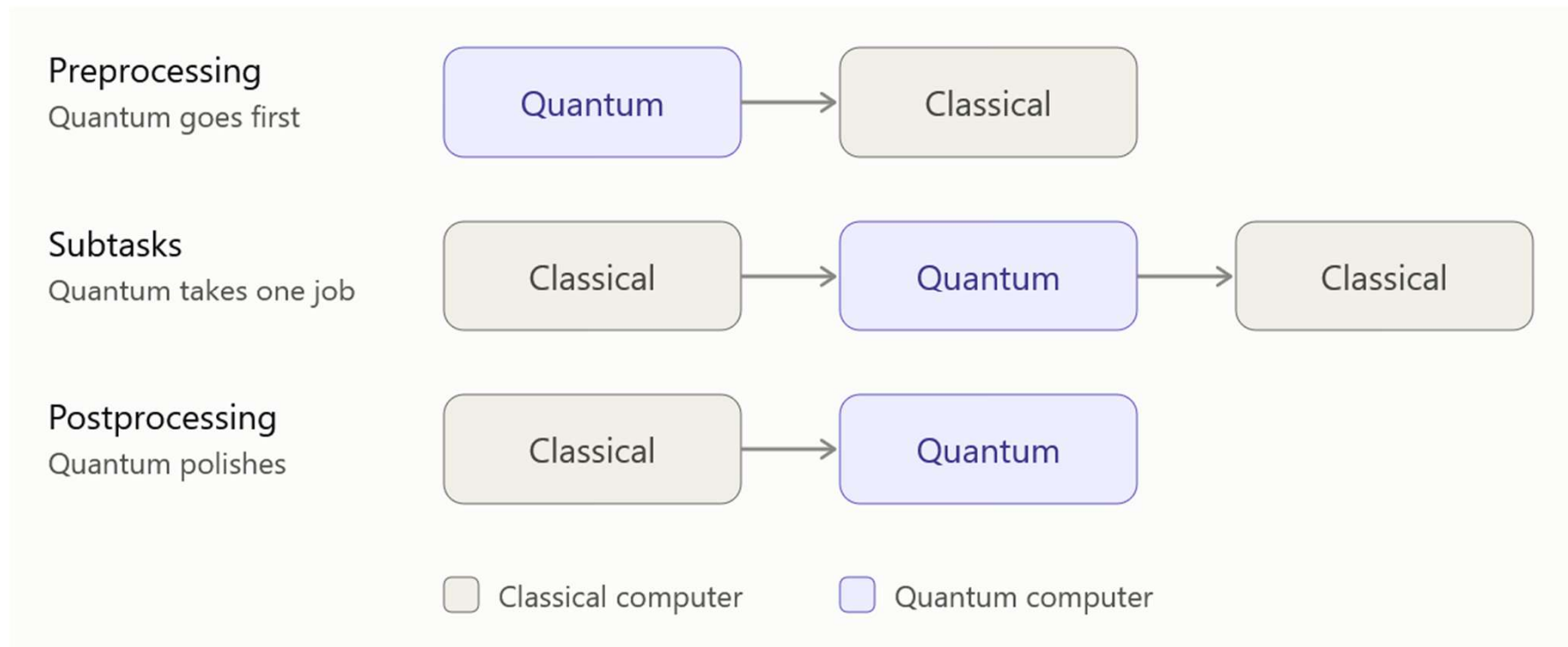


Specialist, not a replacement

- Quantum computers are not better at solving every problem compared to classical computers
- Quantum computers will not replace classical computers
- Quantum computers will work alongside classical computers
 - Hybrid quantum-classical workflows
 - Quantum computer can be used in different parts of the workflow



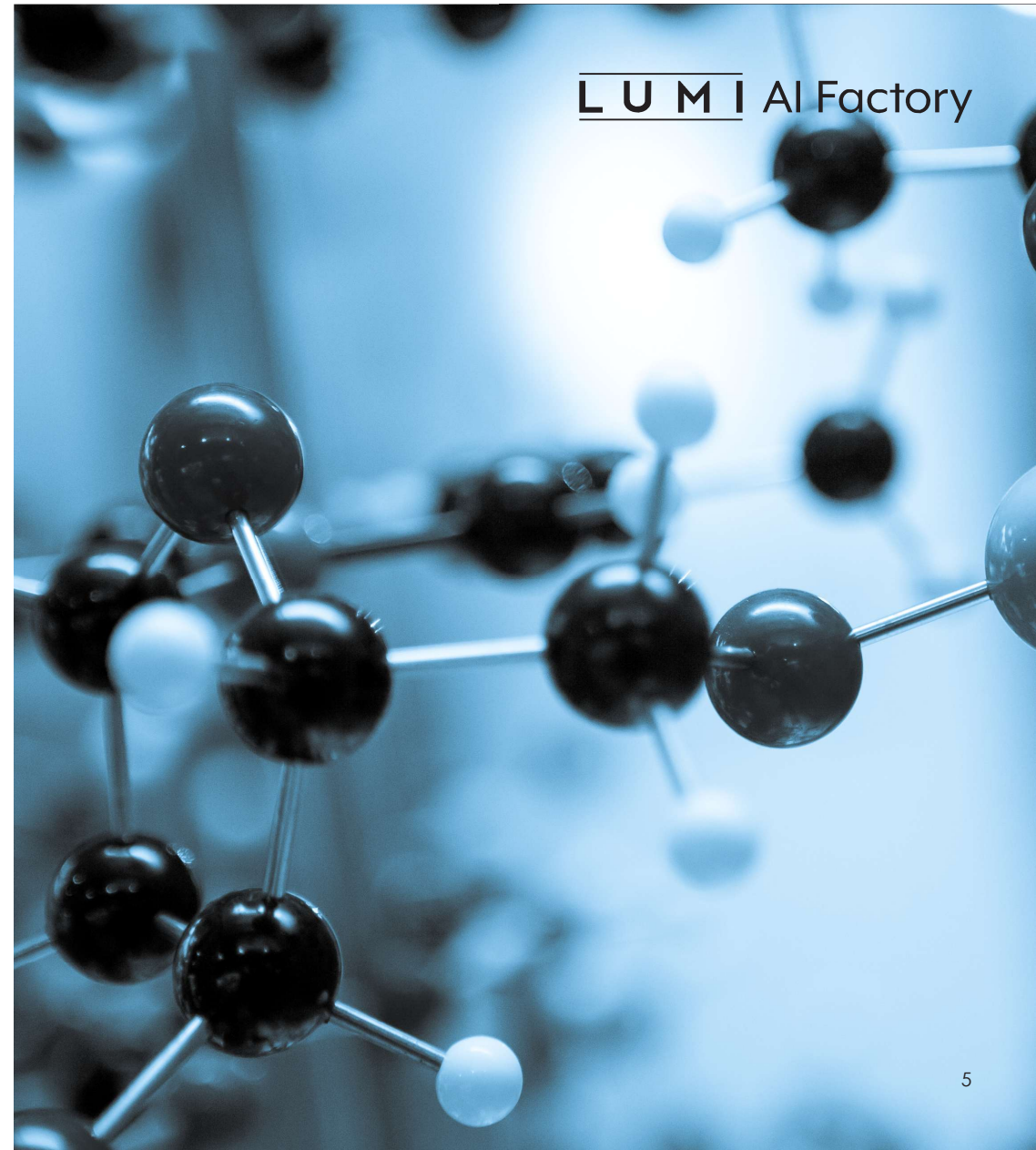
Where Quantum Fits in a Hybrid Workflow



Quantum Computers for Simulating Nature

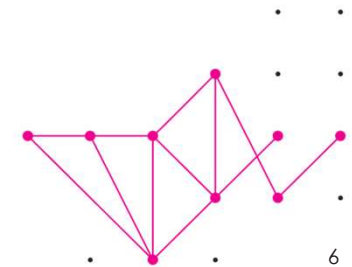
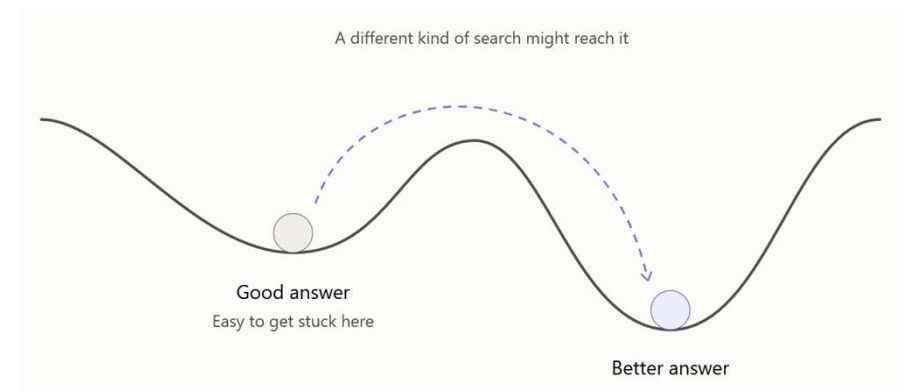
- Natural job for a quantum computer is to model systems that are quantum by nature
- Many problems boils down to the same issue; the cost of classically describing a quantum systems grows exponentially with its size
- Hope to tackle:
 - Explore more efficient catalysts
 - Help with simulating molecules that are hard for classical methods
 - Understand how drugs bind to proteins

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Quantum Computers for Optimization

- The number of possible solutions grows exponentially with the problem size
- Quantum computer searches the landscape of possible solutions differently, hence it might be able to find a better answer
- QAOA and quantum annealers have been developed to solve optimisation problems
- Hope to tackle:
 - Planning delivery routes
 - Scheduling airline crews
 - Balancing electricity supply and demand



Quantum Machine Learning

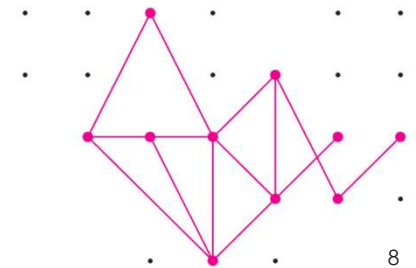
- Machine learning finds patterns in data and the interest in quantum machine learning is whether a quantum computer can find the patterns better or faster
- The cost of loading classical data onto the quantum computer can be quite high
 - Quantum machine learning methods might be better suited for quantum data
- QML might be beneficial for small data sets
- Hope to tackle:
 - Predicting properties of molecules and materials
 - Detecting fraud in financial data

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Quantum Beyond Just Computing

- Random number generation
 - Measuring a quantum system gives a result that is random by the laws of physics
- Quantum sensing
 - Quantum devices are very sensitive to environmental noise
- Exploring physics phenomena
 - Time crystals
 - Particle detection



Reality Check



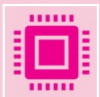
Current-day quantum computers are still small and noisy which affects the performance of quantum algorithms when they are run on real quantum hardware



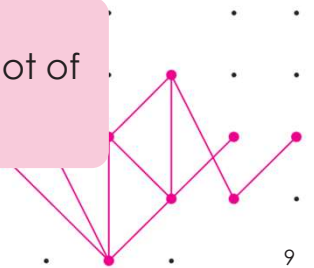
Much of the demonstrations of using quantum algorithms are still at the PoC stage and many of the problems are solvable with classical methods



There has been some instances where some quantum methods have in some sense performed better than classical methods, but later improved classical methods have beat these



Quantum computers and quantum computing still offer many possibilities and they have a lot of potential. Also the industry is evolving fast



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